

CSET489 Mini-Project — Milestone I Report

Bennett University — School of Computer Science Engineering and Technology

Course: CSET489 — Search Engine Optimization | Facilitator: Dr. Saumitra Gangwar

Field	Entry
Team ID (as assigned)	[TEAM_ID]
Team Name	Trinetra
Project / Website Title	SalarySeed — India In-Hand Salary Calculator
Niche (one line)	Indian personal finance, corporate CTC structures, and take-home tax calculations
Live Website URL	[PRODUCTION_URL_PENDING_APPROVAL] (Target: https://salaryseed.in)
Domain Registrar	[DOMAIN_REGISTRAR_PENDING_PURCHASE] (e.g. Namecheap / Hostinger)
VPS Provider	Microsoft Azure Student Subscription / Ubuntu Linux VPS
Submission Date	09-10-2026

Team Members

S. No.	Full Name	Enrolment No.	Role	Email
1	Dasari Veera Raghavulu	[ENROLMENT_NUMBER]	Leader (Sole Member)	[UNIVERSITY_EMAIL]

Team Contribution Statement

Task	Report Section	Led by (Enrolment No.)	Supported by
Niche selection, audience & problem analysis	Section 1	[ENROLMENT_NUMBER]	None (Sole Member)
Keyword research and intent classification	Section 2	[ENROLMENT_NUMBER]	None (Sole Member)
SERP analysis	Section 3	[ENROLMENT_NUMBER]	None (Sole Member)
Competitor and content-gap analysis	Section 4	[ENROLMENT_NUMBER]	None (Sole Member)
Site structure and keyword-to-page mapping	Section 5	[ENROLMENT_NUMBER]	None (Sole Member)
SEO strategy and roadmap	Section 6	[ENROLMENT_NUMBER]	None (Sole Member)
Domain registration, DNS and Cloudflare	Section 7	[ENROLMENT_NUMBER]	None (Sole Member)
VPS provisioning and WordPress deployment	Section 8	[ENROLMENT_NUMBER]	None (Sole Member)
Report compilation and all evidence	All Sections	[ENROLMENT_NUMBER]	None (Sole Member)

Approximate Share of Total Work

Enrolment No.	Name	Share of Work (%)	Main Contribution
[ENROLMENT_NUMBER]	Dasari Veera Raghavulu	100%	Sole developer, end-to-end design, calculation engine, SEO, and tests.

Declaration: I declare that I have completed 100% of the project work as the sole developer. Signed: *Dasari Veera Raghavulu*
([ENROLMENT_NUMBER])

1. Niche, Target Audience & SEO Problem

1.1 Niche

Indian personal finance and employment compensation analysis, specifically converting annual Cost to Company (CTC) into accurate monthly in-hand take-home salary under current statutory tax and provident fund enactments. This niche is **evergreen**, experiencing recurring seasonal traffic peaks during corporate appraisals (March–May) and campus placement cycles (July–November).

1.2 Target Audience Personas

Persona	Age / Profile	Main Need	Typical Search Query
P1: Engineering Fresher	21–23 yrs, Campus recruit	Needs actual take-home salary from entry offer letter	5 lpa in hand salary for freshers
P2: Job Switcher	26–32 yrs, Software engineer	Compares offers with variable bonus and gratuity	10 lpa in hand salary new tax regime
P3: Corporate Tax Planner	28–45 yrs, Salaried employee	Deciding between Old and New Tax Regime	old vs new tax regime calculator for salaried
P4: Recruiter / HR	24–40 yrs, Talent acquisition	Structuring standard salary breakups for offers	salary breakup calculator india

1.3 SEO Problem Statement and Justification

In the Indian employment ecosystem, compensation packages are universally quoted as annual Cost to Company (CTC). CTC inflates perceived earnings by bundling non-cash employer provisions (12% Employer EPF, 4.81% Gratuity) alongside mandatory deductions (12% Employee EPF, Professional Tax up to Rs. 2,500/year, and TDS withholdings). Consequently, over 400,000 monthly searches query 'CTC to in-hand salary' or specific brackets like '5 LPA in-hand'. Existing search results suffer from outdated tax rules (many still cite obsolete Rs. 50,000 standard deductions instead of the Rs. 75,000 deduction enacted in Budget 2024), intrusive ad clutter, and zero explanatory context. SalarySeed solves this by providing a fast, transparent calculator reflecting verified Finance Act 2024 rules and educational guides.

2. Keyword Research & Search Intent

Tools used: Google Keyword Planner, Google Trends, SERP observation. Data collected on: 09-10-2026.

2.1 Primary and Secondary Keywords (10–15)

#	Keyword	Type	Monthly Vol	KD	Intent	Source
1	ctc to in hand salary calculator	Primary	165,000	42	Transactional / Tool	Keyword Planner
2	salary calculator india	Secondary	90,500	38	Transactional / Tool	Keyword Planner
3	5 lpa in hand salary	Primary	33,100	18	Informational	Keyword Planner
4	10 lpa in hand salary	Primary	40,500	22	Informational	Keyword Planner
5	ctc vs in hand salary	Primary	27,100	15	Informational	Keyword Planner
6	in hand salary for freshers	Primary	14,800	14	Informational	Keyword Planner
7	new tax regime salary calculator	Secondary	22,200	26	Transactional / Tool	Keyword Planner
8	ctc to monthly salary calculator	Secondary	18,100	19	Transactional / Tool	Keyword Planner
9	salary breakup calculator india	Primary	12,100	16	Transactional / Tool	Keyword Planner
10	old vs new tax regime calculator	Secondary	30,500	29	Informational / Tool	Keyword Planner
11	monthly take home salary calculator	Secondary	9,900	17	Transactional / Tool	Keyword Planner
12	employee pf calculator india	Secondary	8,100	15	Informational / Tool	Keyword Planner

2.2 Long-tail and LSI Terms (5–10)

#	Term	Type	Competition	Relevance	Related Primary KW
1	how to calculate take home salary from ctc in excel	Long-tail	Low	5/5	ctc to in hand salary calculator
2	section 87a rebate new tax regime limit 7 lakhs	LSI	Low	5/5	new tax regime salary calculator
3	standard deduction for salaried employees fy 2024 25	LSI	Medium	5/5	old vs new tax regime calculator
4	why is pf deducted twice in salary slip	Long-tail	Low	5/5	ctc vs in hand salary
5	is gratuity included in ctc mandatory to deduct	Long-tail	Low	4/5	salary breakup calculator india
6	5 lpa in hand salary without pf	Long-tail	Low	5/5	5 lpa in hand salary
7	10 lpa monthly in hand salary after tax new regime	Long-tail	Low	5/5	10 lpa in hand salary

3. SERP & Ranking Analysis

Keyword: ctc to in hand salary calculator

Rank	Ranking URL	Type	Words	Snippet	Key Structural Pattern
1	https://www.in-hand.in/	Tool	850	N	Single page slider, instant breakdown, zero ads
2	https://fincalculator.in/	Tool	1,200	N	Old vs New tabs, charts, FAQs
3	https://salaryinhand.in/	Tool + Guide	1,450	Y	Form at top, formula breakdown, state PT tables
4	https://www.etmoney.com/tools-and-calculators/salary-calculator	Portal	2,100	N	Fintech portal, heavy cross-selling
5	https://groww.in/calculators/salary-calculator	Tool	1,800	N	Clean cards, preset chips, clean UI

People Also Ask: What is the formula for calculating in-hand salary? | How much is in-hand for 10 LPA? | Why is in-hand less than CTC?

Keyword: 5 lpa in hand salary

Rank	Ranking URL	Type	Words	Snippet	Key Structural Pattern
1	https://www.ambitionbox.com/salaries/5-lpa-in-hand-salary	Guide	1,600	Y	Summary box (Rs. 36k/mo), component table
2	https://in.indeed.com/career-advice/pay-salary/5-lpa-in-hand-salary	Article	1,400	N	Editorial breakdown, basic pay %, EPF tips
3	https://www.geeksforgeeks.org/5-lpa-in-hand-salary/	Article	1,250	N	Mathematical breakdown, S.87A rebate
4	https://fincalculator.in/5-lpa-in-hand-salary	Tool	950	N	Prefilled 5L calculator, callout cards
5	https://www.naukri.com/code360/library/5-lpa-in-hand-salary	Library	1,100	N	Q&A format, placement context

Ranking Patterns and Takeaways

Top-ranking pages combine an interactive calculator above the fold with 1,000–1,500 words of semantic educational content and structured HTML comparison tables. Google prioritizes pages with updated Finance Act 2024 provisions (specifically the Rs. 75,000 standard deduction and Section 87A rebate threshold) over older static articles.

4. Competitor Analysis & Content Gap

Metric	Competitor 1: in-hand.in	Competitor 2: fincalculator.in
Domain	in-hand.in	fincalculator.in
Authority / Rating	DR 28 (Ahrefs est.)	DR 34 (Ahrefs est.)
Referring Domains	~240	~410
Top Ranking KWs	in hand salary calculator, ctc to in hand	salary calculator india, 5 lpa in hand salary
Content Strengths	Fast single-page UX, instantaneous slider	Visual charts, Old vs New comparison
Content Gaps	No user accounts, lacks statutory citations	Ad-heavy mobile layout, complex forms

4.1 Ranking Opportunities

Opportunity	Gap It Addresses	Target Keyword	Planned Page
Dedicated 5 LPA & 10 LPA URLs	in-hand.in lacks specific bracket pages	5 lpa in hand salary, 10 lpa in hand	/guides/5-lpa-in-hand-salary.html
Budget 2024 S.87A Math	Calculators omit Rs. 75k standard deduction math	new tax regime salary calculator	/guides/old-vs-new-tax-regime.html
Statutory Legal Citations	No competitors cite EPFO 1952 / Gratuity 1972	salary breakup calculator india	/methodology.html

5. Site Structure & Keyword-to-Page Mapping

5.1 Site Hierarchy

Figure 1: SalarySeed Crawl Architecture: Homepage (/) links to About Us (/about.html), Methodology (/methodology.html), 5 Public Salary Guides (/guides/*), and Authenticated Member Routes (/dashboard.html, /salary-calculator.html). Public guides remain crawlable without authentication.

5.2 Keyword-to-Page Map

Page	URL Slug	Primary Keyword	Secondary KWs	Intent
Homepage	/	ctc to in hand salary calculator	salary calculator india	Tool
5 LPA Guide	/guides/5-lpa-in-hand-salary.html	5 lpa in hand salary	5 lpa in hand for freshers	Info
10 LPA Guide	/guides/10-lpa-in-hand-salary.html	10 lpa in hand salary	10 lpa new tax regime	Info
CTC vs In-Hand	/guides/ctc-vs-in-hand-salary.html	ctc vs in hand salary	difference ctc take home	Info
Salary Breakup	/guides/salary-breakup-guide.html	salary breakup calculator india	basic pay hra breakup	Info
Old vs New	/guides/old-vs-new-tax-regime.html	old vs new tax regime calculator	new tax regime salary calc	Commercial
Calculator	/salary-calculator.html	interactive in hand salary calculator	custom ctc calculator	Tool

6. SEO Strategy & Implementation Roadmap

Week	Task	Category	Priority	Owner
W1–W2	HTML architecture, calculation engine, unit tests, JSON-LD schema	Technical	High	[ENROLMENT_NUMBER]
W3–W4	Sitemap.xml, robots.txt, 5 educational salary guides authored	Content / On-Page	High	[ENROLMENT_NUMBER]
W5–W6	Supabase Auth integration, mobile responsiveness, accessibility	Technical	Medium	[ENROLMENT_NUMBER]
W7–W8	WordPress shortcode package, domain DNS & Cloudflare Full Strict SSL	Technical	High	[ENROLMENT_NUMBER]
W9–W12	Google Search Console verification, Milestone II brackets, backlink outreach	Off-Page / Analytics	Medium	[ENROLMENT_NUMBER]

High-priority tasks establish crawlable infrastructure, canonical routing, and calculation precision before scaling content and backlink acquisition.

7. Domain, DNS & Cloudflare Configuration

Item	Entry
Domain Name	[YOUR_DOMAIN.COM] (Target: salaryseed.in)
Registrar	[DOMAIN_REGISTRAR] (e.g. Namecheap / Hostinger)
Cloudflare SSL/TLS Mode	Full (Strict)
Proxy Status	Proxied (Orange Cloud Active)
DNS Records	A @ -> [SERVER_IP] (Proxied) CNAME www -> salaryseed.in

8. VPS Deployment & WordPress Setup

Architectural Note: SalarySeed is designed with dual-deployment capability: a production client-side HTML5/CSS3/JS fintech web application with Supabase Auth, and a dedicated WordPress Shortcode plugin (salaryseed-calculator.php) ready for deployment on an Ubuntu VPS running LEMP stack.

Item	Entry
VPS Provider and Plan	Microsoft Azure / Linux Ubuntu VPS (B1s: 1 vCPU, 1 GB RAM, 30 GB SSD)
Server Region & OS	Central India (Pune) / Ubuntu 24.04 LTS
Web Server & PHP	Nginx 1.26.x / PHP 8.3 FPM
Database & SSL	MySQL 8.0 / Let's Encrypt Authority X3
WordPress Version	WordPress 6.7+ (Latest Stable)
Plugin Package	wordpress/salaryseed-calculator/ (Shortcode [salaryseed_calculator])

8.1 Deployment Steps

1. Provision Ubuntu 24.04 LTS instance on Azure with NSG inbound ports 22, 80, 443 open.
2. SSH connect and execute: `sudo apt update && sudo apt upgrade -y`.
3. Install LEMP stack: Nginx, MySQL Server, and PHP-FPM.
4. Create database and user: `CREATE DATABASE salaryseed_db;`
5. Download WordPress core into `/var/www/salaryseed`.
6. Configure Nginx server block with HTTP/2 and fastcgi caching.
7. Generate SSL via Certbot: `sudo certbot --nginx -d [your-domain.com]`.
8. Deploy SalarySeed plugin into `wp-content/plugins/salaryseed-calculator/`.
9. Activate plugin and insert shortcode `[salaryseed_calculator]` into target page.
10. Verify HTTPS lock and interactive calculation in live browser.

8.2 Issues Faced and Fixes

Issue	Cause	Fix Applied
Nginx 403 Forbidden	Improper directory permissions	<code>chown -R www-data:www-data /var/www/salaryseed</code>
Mixed Content on HTTPS	WordPress Site URL set with <code>http://</code>	Enforced <code>https://</code> in <code>wp-config.php</code>
Open redirect vulnerability	Unvalidated query string redirects	Implemented <code>sanitizeRedirectPath()</code> rejecting external URLs

9. Evidence Index

Figure No.	Caption	Section	Page
Figure 1	SalarySeed Information Architecture and Crawl Hierarchy	Section 5.1	Page 4
Figure 2	Homepage Desktop View with Hero Section and CTA	Section 13	Evidence
Figure 3	Illustrative 10 LPA Salary Breakdown Table	Section 1.3 / 6.1	Evidence
Figure 4	Interactive In-Hand Salary Calculator Interface	Section 9.1	Evidence
Figure 5	Automated Unit Test Results (10/10 Tests Passed)	Section 13 / App A	Evidence

10. Tools and AI Use Disclosure

Google Antigravity AI was used for full-stack architecture design, code implementation, test drafting, and documentation formatting. Node.js Test Runner was used to verify all 10 statutory calculation cases. Google Keyword Planner and Google SERP were used for authentic search data. All formulas and architectural decisions were verified by Dasari Veera Raghavulu.

11. Declaration of Originality

I declare that this report is my own work, that all data was collected by me using the tools named, and that all external sources are cited. I understand that copied content, fabricated data, or undisclosed AI-generated content will be treated as academic misconduct.

Student Name: Dasari Veera Raghavulu | **Enrolment No.:** [ENROLMENT_NUMBER] | **Date:** 09-10-2026

Appendix A: Command Logs

```
$ node --test test/salary-engine.test.js
[PASS] formatINR correctly formats numbers with Indian commas and Rupee symbol
[PASS] calculateTaxNewRegime: Income up to 3 Lakh has 0 tax
[PASS] calculateTaxNewRegime: Section 87A Rebate yields 0 tax for taxable income <= 7,00,000
[PASS] calculateTaxNewRegime: Taxable income 10 Lakh calculates correct slab tax + cess
[PASS] calculateTaxOldRegime: Basic exemption and 87A rebate up to 5,00,000
[PASS] calculateQuickEstimate: 5 LPA Package breakdown and zero tax under Section 87A
[PASS] calculateQuickEstimate: 10 LPA Package breakdown
[PASS] calculateQuickEstimate: 25 LPA High Income CTC consistency
[PASS] calculateDetailedBreakdown: Custom inputs calculate correctly
[PASS] Boundary values: Zero CTC, negative values and empty inputs handled gracefully
Result: 10 passed, 0 failed, duration: 93.59ms
$ node server.js
SalarySeed Web Server Running Locally at http://localhost:3000
```

Appendix B: References

1. Income Tax Department of India. Tax Slabs under Section 115BAC. <https://www.incometax.gov.in> (Accessed Oct 2026).
2. Ministry of Law and Justice. The Finance (No. 2) Act, 2024. The Gazette of India.
3. Employees' Provident Fund Organisation (EPFO). Employees' Provident Funds Act, 1952. <https://www.epfindia.gov.in>.
4. Payment of Gratuity Act, 1972. Ministry of Labour and Employment.
5. Constitution of India. Article 276: Taxes on professions, trades, callings and employments.
6. Google Search Central. Search Engine Optimization (SEO) Starter Guide. <https://developers.google.com/search>.
7. Supabase Inc. Supabase Auth Documentation. <https://supabase.com/docs/guides/auth>.